Modeling and Suppression of Common-Mode Current Resonance in an All-in-One Motor Drive Module of Electric Vehicle

Zhaocheng Zhong, Zili Zhu, Jing Sun, and Henglin Chen

***Abstract*—In all-in-one motor drive module of electric vehicle, common-mode (CM) current resonance seriously affects the electromagnetic compatibility (EMC) performance of the motor drive module. To achieve low-cost and effective elimination of CM current resonance, it is necessary to construct an all-in-one motor drive module electromagnetic interference (EMI) model and find out the relevant influencing factors. Due to complex structures, it is difficult to extract the parasitic parameters of dc busbar and bus capacitor in the all-in-one module. This article proposes an extraction method for stray inductance of dc busbar based on current-oscillation frequency, and builds a high-frequency model of multiport bus capacitor based on partial element equivalent circuit and genetic algorithm method. Accordingly, an overall EMI coupling model of the all-in-one module is constructed. The accuracy of the EMI model is verified by comparing simulation results with measurement results. Based on the EMI model, sensitivity analysis is performed to identify the primary CM resonant paths, and resonance mechanism of CM current of the system is analyzed. To eliminate CM current resonance, an EMC quantitative design method is developed. Experiments are performed to validate the effectiveness of the EMC quantitative design method. The problem of CM current resonance in the all-in-one motor drive module is solved effectively.**

***Index Terms*—All-in-one motor drive module, electromagnetic interference (EMI) model, parasitic parameters, resonance suppression.**

I. Introduction

Electromagnetic compatibility (EMC) of electric vehicle motor drive modules is always a primary research focus for major automobile manufacturers [1], [2], [3]. Initially, the EMC design of the automobile motor drive module was relatively straightforward [4], [5]. However, in pursuit of enhanced efficiency, cost reduction, and minimized system volume, automobile manufacturers began integrating inverter, control, charging, and other functions into all-in-one motor drive modules [6], [7], [8]. The high voltage part of the all-in-one motor drive module is composed of line impedance stabilization network (LISN), dc cable, electromagnetic interference (EMI) filter, multiport bus capacitor (MPBC), fast-charging cable, three-phase inverter, and motor, as shown in Fig. 1. This integration exacerbates the common-mode (CM) interference issue within the system, particularly due to the resonance of the CM current, which causes the current method test and the single-pole antenna radiation test to be substandard. Currently, few researches were conducted on EMI modeling of all-in-one motor drive modules, and the CM current resonance mechanism of all-in-one motor drive modules remains unknown. Therefore, it is imperative to construct an accurate EMI model of all-in-one motor drive module and explore the mechanism of CM current resonance. Concurrently, the spatial capacity within the module is constrained, thereby limiting the volume of the filter device. To suppress the CM current resonance [9], the EMC quantitative design method of the motor drive module is indispensable [10].

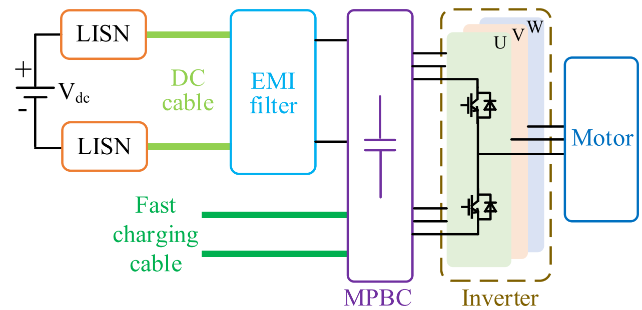

Fig. 1. Schematic diagram of an all-in-one motor drive module.

Many researchers contributed to the modeling of EMI in motor drive modules [11], [12], [13]. However, the focus of these studies is on traditional motor drive modules, with limited attention given to the development of EMI models for all-in-one motor drive modules [14], [15]. At present, two primary challenges exist in the EMI modeling of all-in-one motor drive modules: the precise extraction of the stray inductance of the dc busbar in the inverter and the high-frequency modeling of the MPBC. In the extraction of dc busbar stray inductance, [16] proposed utilizing dynamic waveform of the switching device to obtain the stray inductance of the busbar, which is the traditional time-domain method. In [17], the stray inductance of the dc busbar was extracted by using an applied capacitor to generate the voltage resonance. However, these existing extraction methods

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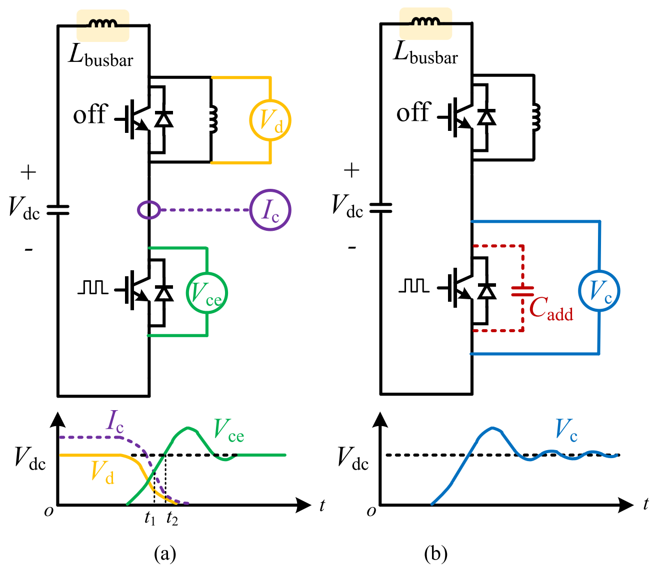

Fig. 2. Conventional stray inductance extraction platform and waveform tested region for dc busbar. (a) Time-domain integration method. (b) Voltage oscillation method.

are cumbersome, and the influence of high-frequency parasitic parameters on the measurement results is not considered. Regarding the high-frequency model of the busbar in the MPBC, scholars suggested employing partial element equivalent circuit (PEEC) method [18] and transformation matrix [19], but did not give a simple extraction method of the embedded capacitors in the MPBC. The references above presented significant research and effective modeling techniques for EMI in motor drive modules.

For the suppression of EMI resonance in motor drive modules, the mechanism of resonance needs to be analyzed. In [20], the reason for the radiation resonance of the motor drive module frequency at 1.36 MHz was analyzed, and the feasibility of the optimization method was verified. In [21], the SPICE equivalent circuit model of the motor drive module was constructed and verified by the input impedance of the model port. Chen et al. [22] delved into the analysis of the self-resonance caused by the EMI filter. Huang and Li [23] discussed the suppression of resonance by asynchronous carriers. Rathnayake et al. [24] presented sensitivity analysis method to find resonant paths. In a word, the aforementioned researches mainly discussed the suppression methods of EMI resonance in motor drive module.

Despite these previous studies on EMI in motor drive modules, there are still some specific problems and difficulties in the extraction of high-frequency parameters and the suppression of resonance. The objects of these previous studies primarily centered on traditional motor drive modules, neglecting the complexities associated with extracting stray inductance parameters of the dc busbar in all-in-one modules, as well as the inconsistent impedance of different ports of the MPBC in EMI modeling. Additionally, the analysis of the resonance mechanism overlooked the impact of certain parasitic parameters. The optimization methods of suppressing EMI lacked quantitative analysis. Therefore, it is necessary to comprehensively investigate the high-frequency parameters extraction method in EMI modeling and a quantitative design approach for resonance suppression.

In order to solve the CM current resonance problem of all-in-one motor drive module accurately and efficiently, this article proposes a novel modeling method to establish an overall EMI coupling model and presents an EMC quantitative design method for resonance suppression for an all-in-one motor drive module.

The rest of this article is organized as follows. In Section II, an extraction method for stray inductance of dc busbar based on current-oscillation frequency and a high-frequency model of MPBC based on partial element equivalent circuit and genetic algorithm (PEEC-GA) method are proposed. According to the proposed methods, an overall EMI coupling model is built. In Section III, based on the EMI model, the sensitivity analysis is performed to identify the primary CM resonant paths, and the mechanism of CM current resonance is further analyzed. In Section IV, a quantitative design method for suppressing EMI resonance is presented and validated by experiment. Finally, Section V concludes this article.

II. EMI Modeling of All-in-One Motor Drive Module

The primary focus is on the analysis of CM current resonance and the conducted interference coupling of the all-in-one motor drive module. Therefore, it is necessary to construct an accurate EMI model with a frequency range from 150 kHz to 30 MHz. Each module in Fig. 1 is individually modeled to construct an overall EMI coupling model of the all-in-one motor drive module. Challenges arise during the EMI modeling process, particularly in the extraction of the stray inductance of the dc bus and addressing the inconsistent impedance of different ports of the MPBC. To mitigate these challenges, two novel methods are proposed. The stray inductance of the dc busbar in the three-phase inverter is extracted by the current-oscillation frequency method, and the MPBC is modeled by the PEEC-GA method.

A. Extraction Method for Stray Inductance of DC Busbar

The MPBC and IGBT module are interconnected through a dc busbar. Direct measurement of the complex dc busbar structure using an impedance analyzer is inconvenient, because it needs time-cost correction work to ensure the accuracy due to the placement and errors from the clip. Therefore, scholars proposed experimental testing methods to obtain the results, as shown in Fig. 2. Commonly employed busbar stray inductance extraction methods mainly include time-domain integration method [16] and voltage oscillation method [17]. Conventional time-domain method utilizes the half-bridge unit in a converter for the pulse generation, as shown in Fig. 2(a). During the test, the voltage between the collector and emitter of the power device will drop or overshoot, which is the theoretical basis of the traditional busbar stray extraction parameter method. However, the measurement accuracy of this method is easily affected by the integral time zone selection (t_1-t_2) in the nonideal switching transient and the synchronization of the test probes. To solve this problem, the voltage oscillation method by adding a resonant capacitor C_{add}

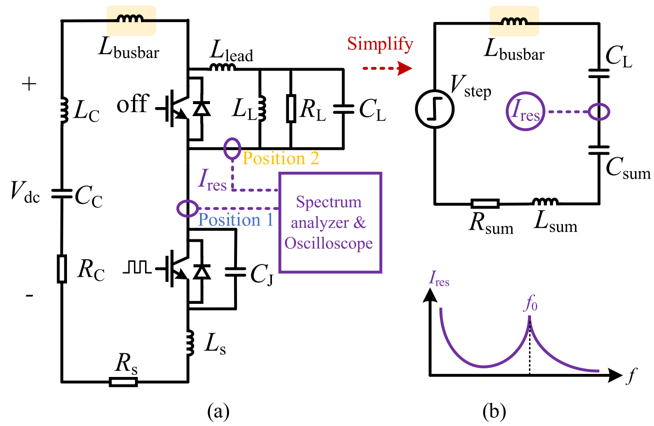

Fig. 3. Proposed extraction method for stray inductance of dc busbar based on current-oscillation frequency. (a) Stray inductance extraction platform. (b) Equivalent circuit and the theoretical spectrum of $I_{\rm res}$.

is proposed, as shown in Fig. 2(b). The stray inductance in the busbar is obtained by the voltage resonance frequency, but the test process of this method is cumbersome and the influence of high-frequency parameters in the circuit is not considered. Additionally, the accuracy of the voltage oscillation method is compromised when the current or the stray inductance of the busbar is low.

As shown in Fig. 3, this article proposes an extraction method for stray inductance of dc busbar based on current-oscillation frequency. Fig. 3(a) shows the measured situation in inductance-dominated application. When performing high-frequency analysis, it is imperative to recognize that the dc capacitor and air-core inductor do not conform to the characteristics of ideal voltage and current sources [25]. More specifically, the dc capacitor can be elucidated as an RLC series high-frequency equivalent circuit, $R_{\rm C}$ is the equivalent series resistance, $L_{\rm C}$ is the equivalent series inductance (ESL), and $C_{\rm C}$ is the equivalent series capacitance. Meanwhile, the selected air-core inductor can be conceptualized as an RLC parallel high-frequency equivalent circuit, $R_{\rm L}$ is the equivalent parallel resistance, $L_{\rm L}$ is the equivalent parallel inductance, and $C_{\rm L}$ is the equivalent parallel capacitance. Other high-frequency parameters need to be considered. $L_{\rm S}$ is the inductance of the power device module, $C_{\rm J}$ is the junction capacitance, $R_{\rm S}$ is the resistance of the resonant loop, $L_{\rm lead}$ is the lead inductance of the air-core inductor, and $L_{\rm busbar}$ is the busbar stray inductance that needs to be extracted. To facilitate the analytical process, the proposal of an equivalent resonance model for the circuit is necessary. The chosen air-core inductor exhibits an equivalent parallel resistance $R_{\rm L}$ of approximately 11 kΩ and an equivalent parallel inductance $L_{\rm L}$ of around 849 μH, which can be reasonably approximated as two open circuits during high-frequency analysis. Furthermore, the switching process of the power device can be reasonably modeled as a step signal $V_{\rm step}$ with an amplitude of $V_{\rm dc}$ [17]. Additionally, $C_{\rm sum}$ denotes the series combination of $C_{\rm C}$ and $C_{\rm J}$, $L_{\rm sum}$ represents the aggregate of $L_{\rm C}$, $L_{\rm lead}$, and $L_{\rm s}$, and $R_{\rm sum}$ encapsulates the summation of $R_{\rm C}$ and $R_{\rm s}$. Consequently, the simplified equivalent circuit in Fig. 3(b) can be obtained.

TABLE I
ELECTRICAL PARAMETERS IN CURRENT OSCILLATION TEST

Parameter	Value	Parameter	Value	Parameter	Value
$R_{\rm C}$	0.06 Ω	$L_{\rm lead}$	0.25 μH	$C_{\rm C}$	393.6 μF
$R_{\rm L}$	11 kΩ	$L_{\rm L}$	849 μH	$C_{\rm L}$	125 pF
$R_{\rm s}$	0.01 Ω	$L_{\rm s}$	5.8 nH	$C_{\rm J}$	36.9 nF
$L_{\rm C}$	43.5 nH				

The step response of $I_{\rm res}$ is expressed by the frequency domain method as

$$I_{\rm res}(s) = \frac{V_{\rm dc}}{s\left[s(L_{\rm busbar}+L_{\rm sum})+\left(\frac{1}{sC_{\rm sum}}+\frac{1}{sC_{\rm L}}\right)+R_{\rm sum}\right]}. \quad (1)$$

According to (1), in the simplified equivalent circuit of Fig. 3(b), the oscillation frequency f_0 observed in $I_{\rm res}$ satisfies the formula of underdamped oscillation in second-order RLC circuit, as presented in

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{(L_{\rm busbar}+L_{\rm sum})(C_{\rm L}//C_{\rm sum})} - \frac{R_{\rm sum}^2}{4(L_{\rm busbar}+L_{\rm sum})^2}}. \quad (2)$$

In this article, $C_{\rm C}$ is a few hundred μF, $C_{\rm J}$ is also a few tens of nF, but $C_{\rm L}$ is 125 pF. Given that the sum impedance of $C_{\rm sum}$ is relatively minor in comparison to $C_{\rm L}$, it can be disregarded. Additionally, the value of $R_{\rm sum}$ is also relatively insignificant, and the latter term in the root of (2) is considerably smaller than the value of the preceding term, thereby allowing for the simplification of (2) to (3)

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{(L_{\rm busbar}+L_{\rm sum})C_{\rm L}}}. \quad (3)$$

The impedance of the dc capacitor and the air-core inductor can be tested using an impedance analyzer, while high-frequency parameters can be derived through a fitting algorithm. The lead inductance of IGBT modules $L_{\rm S}$ can query the manual, and all parameters mentioned above are given in Table I. To prevent unexpected interference with the test current $I_{\rm res}$ from other signals, it is necessary to measure at positions 1 and 2, respectively, to obtain the same current resonance frequency f_0, as shown in Fig. 3(a). The oscillation method is suitable for measuring the stray parameters of complex busbars [17]. The double pulse test [16] was used to extract the parameters of the circuit in Fig. 3. Spectrum analyzer and oscilloscope were utilized to capture and analyze the signals. In the low frequency band, due to the influence of the driving current and the diode reverse recovery current, the amplitude spectrum envelopes of the two positions are different, and the corresponding time domain waveform is also drawn in Fig. 4.

As shown in Fig. 4, the measured spectrum envelopes of current $I_{\rm res}$ at positions 1 and 2 have the same current resonance frequency $f_0 = 24.97$ MHz. According to (3), the $L_{\rm busbar}$ is extracted as 23.46 nH. The calculation results of the tested dc busbar $L_{\rm busbar}$ are presented in Table II. Compared with the results from impedance analyzer, the proposed method is capable for dc busbar with 2.8% relative error, thus confirming the accuracy of the proposed method.

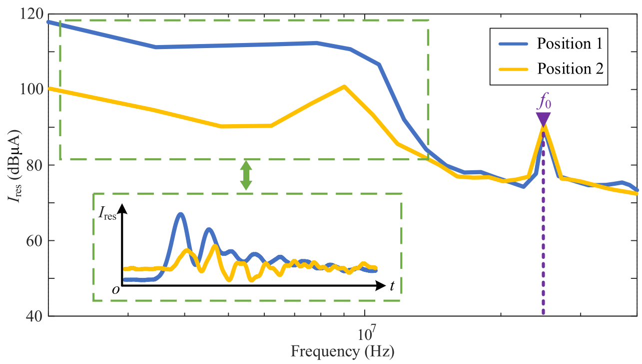

Fig. 4. Spectrum envelope and waveform of $I_{\rm res}$ measured at positions 1 and 2 in stray inductance extraction platform.

TABLE II
EXPERIMENTAL AND IMPEDANCE ANALYZER RESULT OF BUS BAR

Symbol	Experiment	Impedance analyzer	Relative error
$L_{\rm busbar}$	23.46 nH	24.14 nH	2.8%

B. High-Frequency Model of MPBC

MPBC structure is more intricate compared to the traditional bus capacitor, which is typically consisted of multiple embedded capacitors in parallel within the internal dc bus. As illustrated in Fig. 5(a), the MPBC utilized in this article features multiple ports, comprising a dc input terminal and three output terminals. The impedance characteristics of each terminal differ due to the asymmetric distribution of stray inductance in the internal dc busbar. The impact of the stray inductance of its internal dc bus cannot be disregarded. Consequently, an RLC series equivalent circuit cannot be directly employed to characterize the high-frequency model [11], as illustrated in Fig. 5(b). The MPBC module is sealed. The disassembly process would disrupt the original structure, rendering direct impedance measurement of embedded capacitors impractical. In order to ensure that the proposed model holds practical physical significance, the stray inductances need to be considered imperatively. Fig. 5(c) illustrates the 3-D structure of the internal bus of the MPBC used in this article, constructed from copper. The light red portion represents the P-plate, the gray portion signifies the N-plate, and the blue-green portion demarcates the dividing plane of the PEEC method. The PEEC method is suitable for modeling complex structures [26], and according to [18], the capacitor busbar P and N plates can be divided into three segments: $L_{\rm bar1}$; $L_{\rm bar2}$; and $L_{\rm bar3}$. The stray inductances and parasitic capacitances corresponding to each segment and pin are extracted by ANSYS Q3D [27]. The high frequency equivalent model of the capacitor busbar structure is obtained. The equivalent model of each segment is demarcated by a dotted frame in Fig. 5(d). Concurrently, the high-frequency model of 1∼8 pins is replaced by RL series circuits in the circuit structure, with the pin numbers marked in Fig. 5(d).

The conventional PEEC method in [18] is limited to extracting the stray parameters of the internal dc bus. To extract the parameters of the embedded capacitors, a novel method

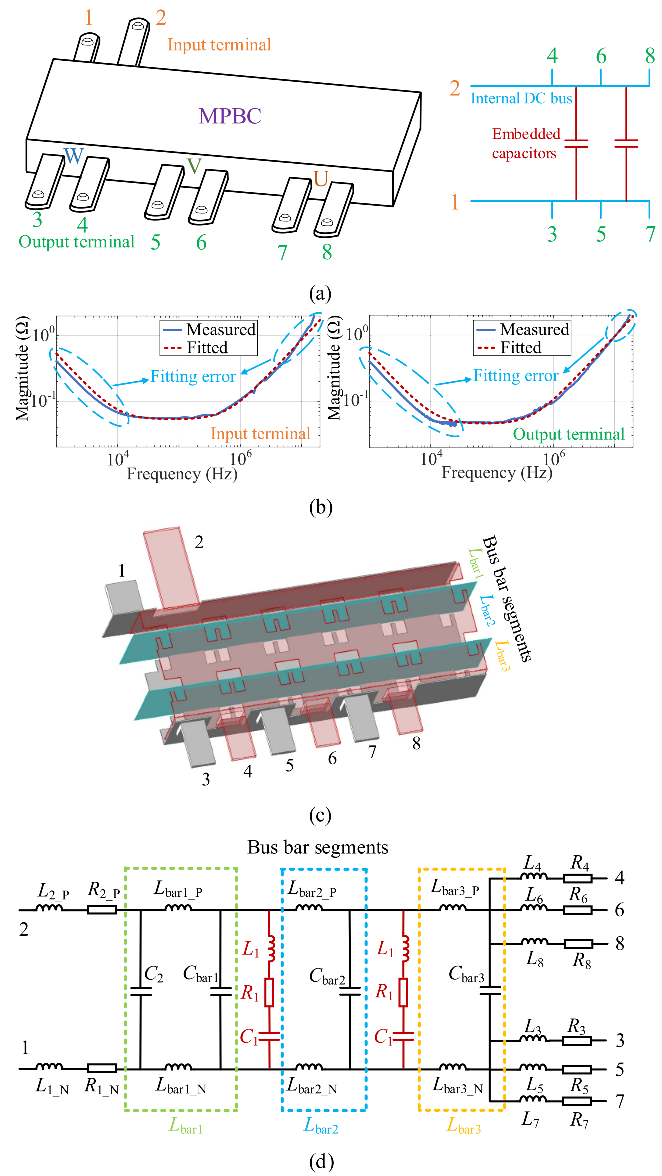

Fig. 5. A high frequency model of MPBC based on PEEC-GA method. (a) Schematic diagram of the MPBC. (b) RLC series equivalent circuit fitting results of input terminal and output terminal. (c) PEEC method segmentation of busbar. (d) Equivalent circuit.

called PEEC-GA is proposed. Based on the results of busbar extraction by PEEC method, the genetic algorithm [11] was used to fit the input impedance measurement results to obtain the high-frequency equivalent model of the embedded capacitors. During the impedance fitting process, the embedded capacitors can be represented by two RLC series structures with consistent high-frequency impedance characteristics, while the corresponding high-frequency equivalent parts are marked red in Fig. 5(d), which are L_1, R_1, and C_1. The fitting results of the specific parameters of the model are given in Table III. The feasibility of the method is demonstrated by comparing the actual measurement results of the output impedance with the model fitting results. During the GA fitting process, the input terminals 1 and 2 were selected, and the output terminals

TABLE III
IMPEDANCE PARAMETERS OF MPBC

Parameter	Value	Parameter	Value	Parameter	Value
L_{1_N}	1.13 nH	L_{2_P}	1.4 nH	L_{bar1_P}	0.79 nH
L_{bar1_N}	0.97 nH	L_{bar2_P}	2.5 nH	L_{bar2_N}	2.7 nH
L_{bar3_P}	0.66 nH	L_{bar3_N}	0.73 nH	L_1	11.5 nH
L_3	5.7 nH	L_4	0.45 nH	L_5	4.9 nH
L_6	0.44 nH	L_7	5.02 nH	L_8	0.44 nH
R_{1_N}	300 μΩ	R_{2_P}	20 μΩ	R_1	92 mΩ
R_3	1.25 mΩ	R_4	0.2 mΩ	R_5	1 mΩ
R_6	0.2 mΩ	R_7	0.9 mΩ	R_8	0.2 mΩ
C_1	198 μF	C_2	5.48 nF	C_{bar1}	1.4 pF
C_{bar2}	1.4 pF	C_{bar3}	148 pF		

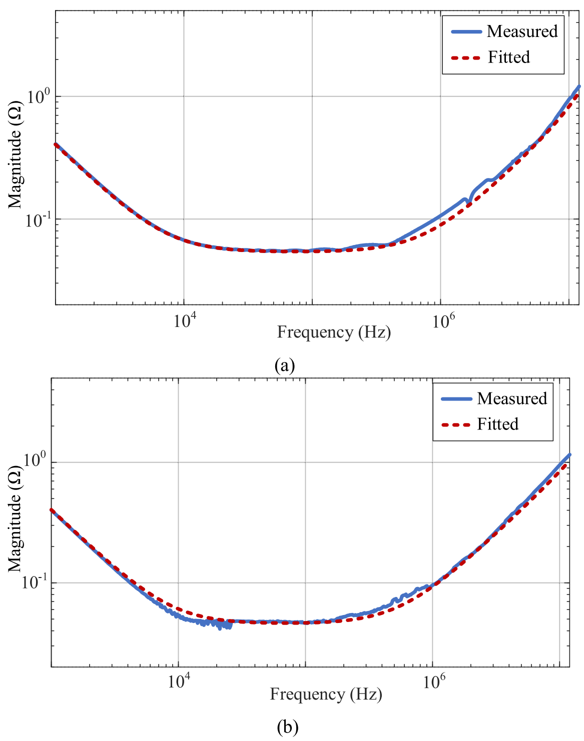

Fig. 6. Comparison of measured and fitted MPBC impedance results by PEEC-GA method. (a) Comparison of measured and fitted input terminal impedance. (b) Comparison of measured and fitted output terminal impedance.

3 and 4 were selected. As shown in Fig. 6, the fitting results of the impedance at the dc input terminal and output terminal agree well with the measurement results respectively, which illustrate the accuracy of the PEEC-GA method for high-frequency modeling of MPBC.

C. Overall EMI Coupling Model

Based on the impedance model built of dc busbar and MPBC, an overall EMI coupling model of the all-in-one motor, drive module is proposed, as illustrated in Fig. 7. Within this module, the inverter is directly linked to the motor through a three-phase busbar, which discharges into the motor module during the modeling process. The dc cable, EMI filter and Motor use impedance analyzer and software to extract parameters [27]. They are divided into CM and differential-mode (DM) parts by algorithm [28]. The corresponding parameter symbols are marked in Fig. 7. Z_{DCM} and Z_{DDM} are CM & DM parts of dc cable. Z_{ECM} and Z_{EDM} are CM and DM parts of EMI Filter. Z_{MCM} and Z_{MDM} are CM & DM parts of motor. The 1–8 ports of the MPBC correspond to the port symbol in Fig. 5. The CM model of the fast-charging cable can be equivalent to an RLC series circuit, which are R_{fc}, L_{fc}, and C_{fc}. The ground capacitances C_{P}, C_{o}, and C_{N} of the IGBT module in inverter are extracted by resonance method [29], culminating in the derivation of a comprehensive CM EMI coupling model. The operating power of the all-in-one motor drive module is 20 kW, and the rated dc voltage is 400 V. Fig. 8 delineates the layout of the actual test arrangement, conforming to the EMI test standard.

The overall CM EMI coupling model was constructed in the simulation software [30], [31]. The frequency domain simulation result of the CM current was obtained and compared with the spectrum envelope of the measurement result, as depicted in Fig. 9. The simulation result is consistent with the measurement result. From the CM current simulation and measurement results in Fig. 9, there are two resonance frequencies of 16 MHz and 28 MHz in the frequency range from 150 kHz to 30 MHz. Therefore, it is imperative to analyze the mechanism of CM current resonance and investigate a quantitative-design suppression method.

III. RESONANCE MECHANISM OF CM CURRENT

To comprehensively assess and effectively suppress the CM EMI of high voltage power supply port, a thorough investigation of the CM current resonance mechanism of the all-in-one motor drive module is necessary. The CM EMI coupling paths in the all-in-one motor drive module are rather complicated. In this article, the frequency of the CM current resonance frequencies in motor drive module are higher than 10 MHz. By directing attention to the distributed parameters related to the resonance circuit within the 10 MHz to 30 MHz frequency range, which can streamline the overall EMI coupling model in Fig. 7.

In Fig. 7, the rapid turn-ON and turn-OFF of the IGBT induce a transition in the ground voltage, resulting in CM interference. Section II obtains the EMI model of the MPBC and the inductance of the dc busbar L_{busbar}, which primarily influence the CM voltage interference source and significantly impact the EMI simulation results detailed in Section II. However, the impedance of the MPBC and the dc busbar has minimal effect in the resonant circuit, resonant path analysis is not harsh on the accuracy of the interference source. Therefore, the bus capacitor to the IGBT module in Fig. 7 can be collectively considered as CM equivalent voltage noise source V_{SCM} [22]. LISN can be perceived as a series composition of C_{LI} and R_{LI}, dc cable can be simplified to an equivalent inductance L_{cable} and ground capacitance C_{cable}. C_{o} is the ground capacitance at the midpoint of the inverter bridge arm. Considering the influence of stray inductance at high frequency, the ground capacitance of the EMI filter can be approximately disregarded and simplified to a CM inductance L_{CM}. L_{bar} denotes the high-frequency attributes of the three-phase busbar linking the inverter and the motor. The CM model of the motor can be roughly simplified to L_{motor}, C_{bm}, and C_{motor}. While ignoring some electrical parameters in the circuit due to their smaller influencing factors. Therefore, the

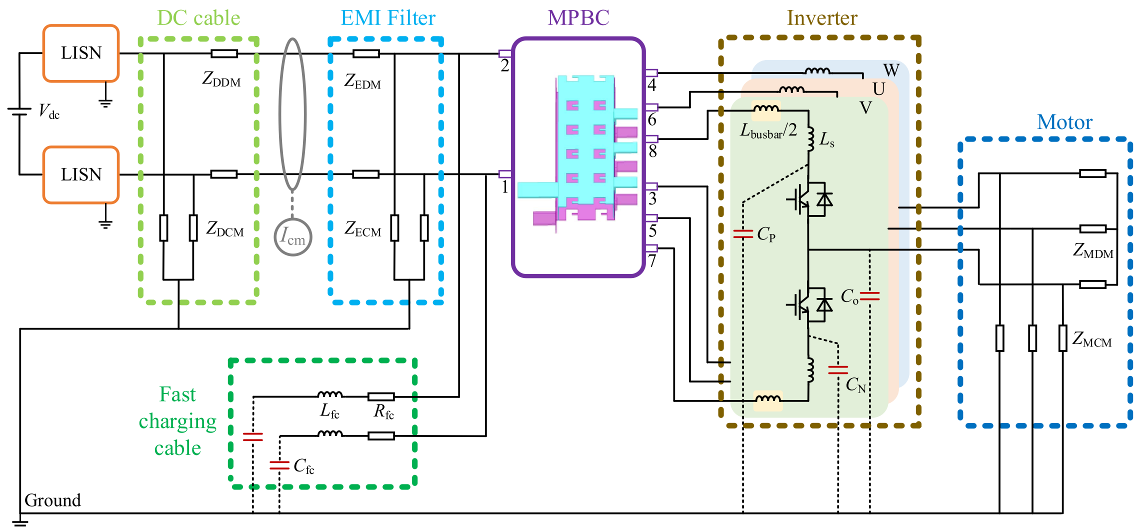

Fig. 7. Overall EMI coupling model of the all-in-one motor drive module.

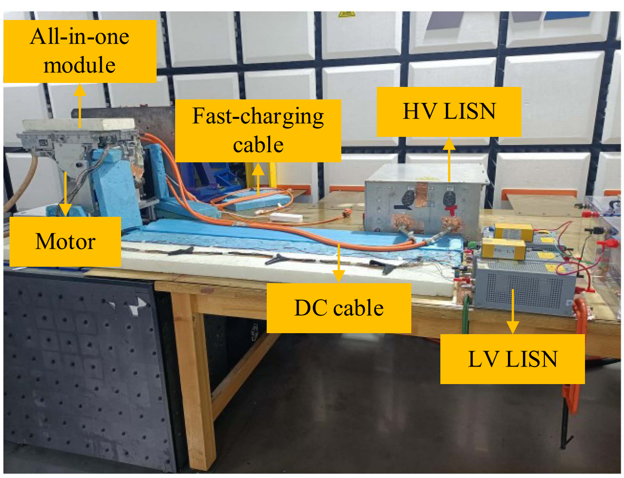

Fig. 8. EMI test arrangement.

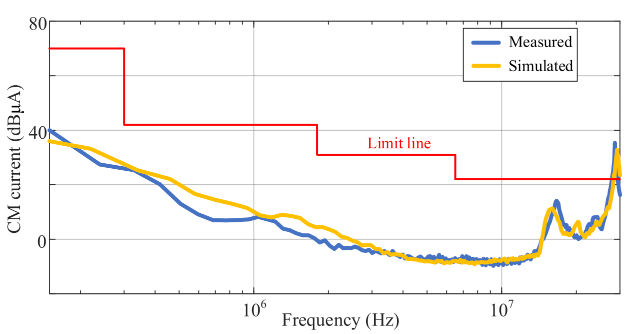

Fig. 9. Comparison of measured and simulated CM current.

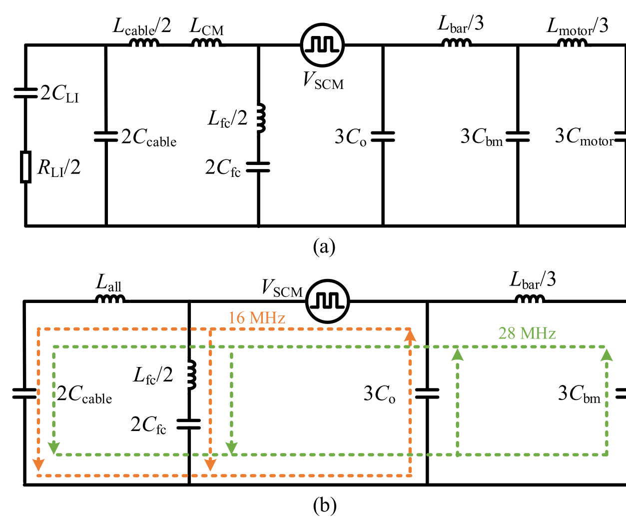

Fig. 10. CM circuit for all-in-one motor drive module. (a) Equivalent CM circuit. (b) Simplified CM circuit.

TABLE IV
ELECTRICAL PARAMETERS OF THE SIMPLIFY EQUIVALENT CIRCUIT

Parameter	Value	Parameter	Value	Parameter	Value
R_{LI}	50 Ω	L_{cable}	1.8 µH	L_{CM}	1.1 µH
L_{fc}	2 µH	L_{bar}	25 nH	L_{motor}	0.6 mH
C_{LI}	0.1 µF	C_{cable}	24.5 pF	C_{fc}	49 pF
C_o	0.82 nF	C_{bar}	30.7 pF	C_{motor}	0.91 nF

overall EMI coupling model can be regarded as an equivalent CM circuit in frequency range from 10 to 30 MHz, as depicted in Fig. 10(a), with specific parameters given in Table IV.

The simulated CM current spectrum of the equivalent CM circuit is depicted in Fig. 11, compared with the result in Fig. 9, which effectively validates the precision of the proposed equivalent CM circuit within the 10 to 30 MHz frequency range. To streamline the analysis of the resonance mechanism, it suffices to concentrate solely on the paths with a frequency range from 10 to 30 MHz. Additionally, the impedance of the motor CM inductor $L_{motor}/3$ is sufficiently large at 16 and 28 MHz, which can be regarded as disconnected.

In this article, sensitivity analysis [24] is performed on the electrical parameters in the equivalent CM circuit to identify the

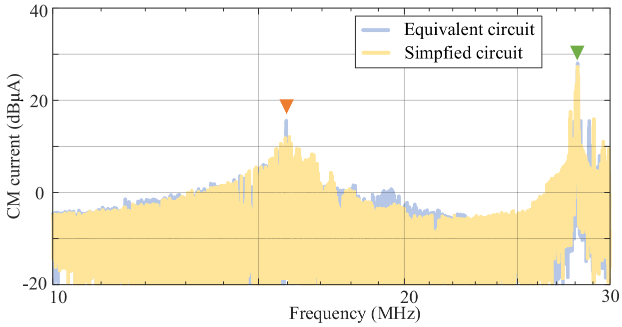

Fig. 11. Simulation spectrums of the equivalent CM circuit and the simplified CM circuit.

key parameters associated with the resonance frequencies and to identify the main CM resonance paths. This approach enables the identification of multiple resonant paths within the circuit simultaneously. Ultimately, the equivalent CM circuit can be simplified. Fig. 10(b) provides a simplified CM circuit where $L_{\mathrm{all}} = L_{\mathrm{cable}}/2 + L_{\mathrm{CM}}$. The simplification process will cause some spectral amplitude deviations [32], [33]. The simulated CM current spectrum of the simplified CM circuit is illustrated in Fig. 11 within the frequency range of 10 to 30 MHz, which agrees well with the simulated spectrum of equivalent CM circuit. This substantiates the applicability of the proposed simplified CM circuit for analyzing the resonance mechanism.

As depicted in Fig. 10(b), two resonance paths emerge at 16 MHz. The CM paths at 16 MHz are represented by orange lines. The CM interference originates from the equivalent CM interference source V_{SCM}, with one path propagating through the fast-charging cable and coupling to the ground via capacitance C_{fc}, and the other path propagating through the dc cable and coupling to the ground through the CM capacitance C_{cable}. Subsequently, interference couples to V_{SCM} through the capacitance C_{o}.

As shown in Fig. 10(b), the CM paths at 28 MHz are represented by green lines. The interference originates from the equivalent CM interference source V_{SCM}, propagating to the dc cable and fast-charging cable, and then coupling back to V_{SCM} through the CM capacitance C_{bm} and C_{o}.

IV. Resonance Suppression

In this article, the EMI of the all-in-one motor drive module did not meet the CISPR25 standard in the frequency range of 150 kHz to 30 MHz. Based on the obtained overall EMI coupling model and resonance mechanism, the existing filter device needs to be optimized to suppress the CM current EMI resonance. Considering design and manufacturing costs, the proposed suppression plan aims to strike a balance between simplicity and effectiveness. Consequently, this article introduces a quantitative design approach that leverages simplified CM circuit for analysis and design, as shown in Fig. 12. The applied CM capacitors C_{Y}, highlighted in red, are strategically positioned at the juncture between the fast-charge cable and the MPBC. Two C_{Y} effectively establish a new path for the CM loop, thereby attenuating the current resonance in the CM path.

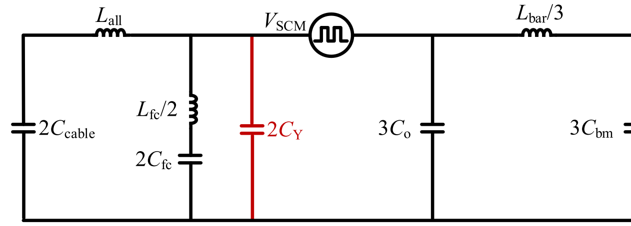

Fig. 12. Simplified CM circuit after connected two C_{Y}.

As long as the insertion loss required by CM current at the resonance frequencies is calculated, the required value of CM capacitor C_{Y} can be obtained. In Fig. 10(b), the calculation of the 16 MHz resonant circuit is shown in (4) and (5), where Z_{Y16} is the parallel impedance of the dc cable and fast-charging cable, and the CM current result is I_{CM16}

$$Z_{\mathrm{Y16}} = \left(sL_{\mathrm{all}} + \frac{1}{2sC_{\mathrm{cable}}}\right) // \left(sL_{\mathrm{fc}}/2 + \frac{1}{2sC_{\mathrm{fc}}}\right) \tag{4}$$

$$I_{\mathrm{CM16}} = \frac{sL_{\mathrm{fc}}/2 + \frac{1}{2sC_{\mathrm{fc}}}}{Z_{\mathrm{Y16}}} \frac{V_{\mathrm{SCM}}}{Z_{\mathrm{Y16}} + \frac{1}{3sC_{\mathrm{o}}}}. \tag{5}$$

Fig. 12 is the simplified CM circuit after install C_{Y}. Z_{Y16C} is the parallel CM impedance of the dc cable, fast-charge cable, and C_{Y}. The CM current result in Fig. 11(a) is expressed as I_{CM16C}, as shown in

$$Z_{\mathrm{Y16C}} = \left(sL_{\mathrm{all}} + \frac{1}{2sC_{\mathrm{cable}}}\right) // \left(sL_{\mathrm{fc}}/2 + \frac{1}{2sC_{\mathrm{fc}}}\right) // \frac{1}{2sC_{\mathrm{Y}}} \tag{6}$$

$$I_{\mathrm{CM16C}} = \frac{\left(sL_{\mathrm{fc}}/2 + \frac{1}{2sC_{\mathrm{fc}}}\right) // \frac{1}{2sC_{\mathrm{Y}}}}{Z_{\mathrm{Y16C}}} \frac{V_{\mathrm{SCM}}}{Z_{\mathrm{Y16C}} + \frac{1}{3sC_{\mathrm{o}}}}. \tag{7}$$

From (4) to (7), it is evident that the calculated CM current insertion loss $\mathrm{IL}_{\mathrm{16M}}$ at 16 MHz is

$$\begin{aligned} \mathrm{IL}_{\mathrm{16M}} &= 20\lg I_{\mathrm{CM16}} - 20\lg I_{\mathrm{CM16C}} \\ &= 20\lg \frac{I_{\mathrm{CM16}}}{I_{\mathrm{CM16C}}}. \end{aligned} \tag{8}$$

As shown in Figs. 10(b) and 12, a simplified CM circuit for the 28 MHz resonance path are depicted. The corresponding insertion loss $\mathrm{IL}_{\mathrm{28M}}$ can be expressed as (9), I_{CM28} represents the CM current before quantitative design. I_{CM28C} represents the result after quantitative design

$$\begin{aligned} \mathrm{IL}_{\mathrm{28M}} &= 20\lg I_{\mathrm{CM28}} - 20\lg I_{\mathrm{CM28C}} \\ &= 20\lg \frac{I_{\mathrm{CM28}}}{I_{\mathrm{CM28C}}}. \end{aligned} \tag{9}$$

After the quantitative design, the magnitude should maintain a 5 dB margin, ensuring that the insertion loss $\mathrm{IL}_{\mathrm{16M}}$ and $\mathrm{IL}_{\mathrm{28M}}$ are at least 20 dB. The 20 dB value is integral to the actual calculation process. Moreover, the EMC design process must account for the influence of leakage current. The collective capacitance of the additional filter Y capacitors should not surpass 90 nF. In

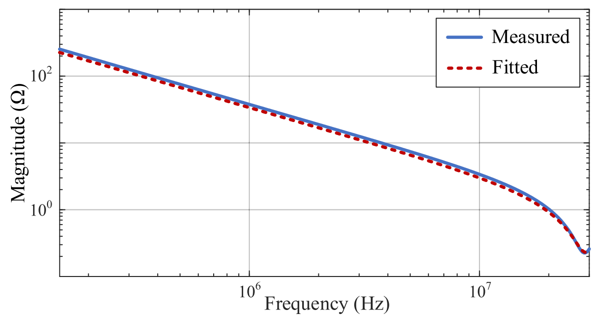

Fig. 13. Comparison of measured and fitted C_{Y} impedance.

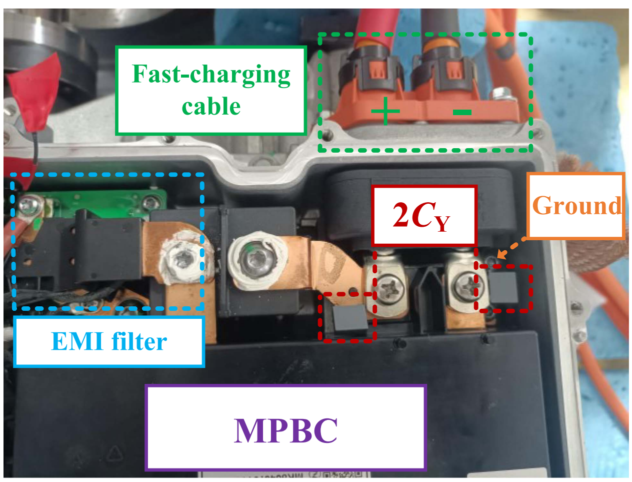

Fig. 14. Schematic diagram of Y capacitors connection.

light of the aforementioned analysis, it can be deduced that the value of C_{Y} should adhere to

$$\begin{cases} \mathrm{IL}_{16\mathrm{M}} > 20\ \mathrm{dB}, f = 16\ \mathrm{MHz} \\ \mathrm{IL}_{28\mathrm{M}} > 20\ \mathrm{dB}, f = 28\ \mathrm{MHz} \\ 2C_{\mathrm{Y}} < 90\ \mathrm{nF}. \end{cases} \tag{10}$$

Applying the data in Table IV into (10), we obtained 8.8 nF $< 2C_{\mathrm{Y}} <$ 90 nF. To minimize leakage current during the design process, C_{Y} takes the value of 4.7 nF. The impedance of C_{Y} is calculated by genetic algorithm [11], and its ESL is obtained as 6 nH, as shown in Fig. 13. The ESL of 2C_{Y} is 3 nH in Fig. 12, so their impedance can be approximately ignored. If the ESL of the Y capacitor reaches ten or dozens of nH, the ESL of Y capacitor needs to be considered during the analysis process. Add 2C_{Y} to the EMI coupling model of Fig. 7. The simulation result indicates that the CM current EMI is lower than the emission limit, as depicted by the blue line in Fig. 16(a). Subsequently, Y capacitors are integrated at the connection between the fast-charging cable and MPBC, as shown in Fig. 14. The low-frequency radiation of the motor drive module is mainly generated by CM current of the cable [20], [34], [35]. Single-pole antenna radiation was carried out to verify the effectiveness of the suppression method, as shown in Fig. 15. The EMI experiments were conducted, as illustrated in Fig. 16.

Upon comparison with Fig. 9, the two CM current resonance frequencies of the all-in-one motor drive module are attenuated. After using the EMC quantitative design scheme, the radiation is also suppressed. Conduction and radiation EMI of the system

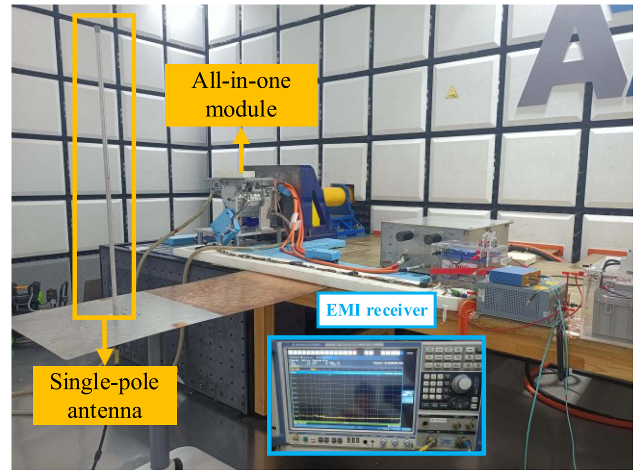

Fig. 15. Radiated EMI test arrangement.

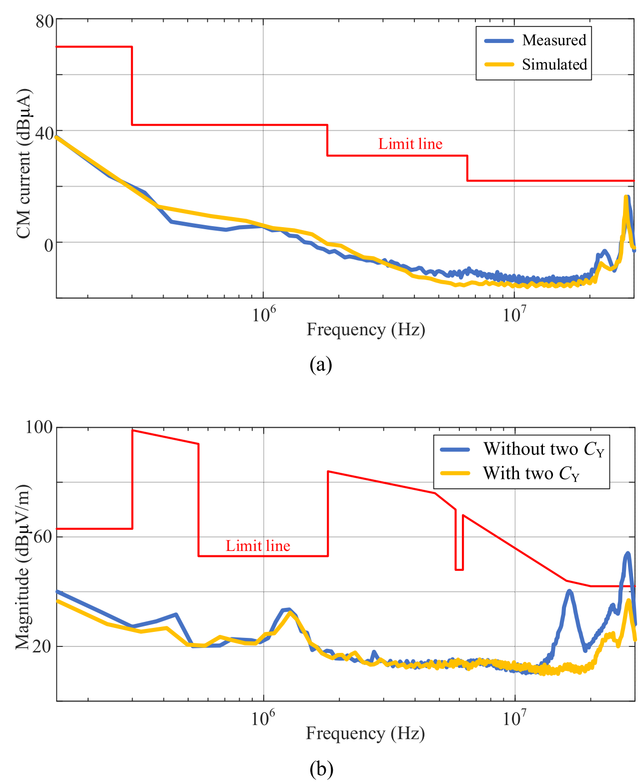

Fig. 16. Spectrum envelopes of simulation and measurement results. (a) Comparison of measured and simulated CM current after using the EMC quantitative design scheme. (b) Comparison of the single-pole antenna radiation measurement results between before two C_{Y} are connected and after two C_{Y} are connected.

are suppressed below the standard limit. Therefore, the EMC quantitative design scheme conforms to the CISPR 25 standard in frequency range from 150 kHz to 30 MHz, validating that the proposed quantitative design suppression method is effective.

V. Conclusion

In all-in-one motor drive module of electric vehicle, CM current resonance seriously affects the EMC performance of the module. This article illustrates the resonance mechanism in the all-in-one motor drive module by constructing an overall EMI coupling model. To accurately evaluate the interference level of all-in-one motor drive module, an extraction method for stray inductance of dc busbar based on current-oscillation frequency and a high-frequency model of MPBC based on PEEC-GA method are proposed. The accuracy of the model is verified by comparing the simulation result of CM current with the measurement result. Based on the EMI model, the sensitivity analysis is performed to identify the primary CM resonant paths. Then, resonance mechanism of CM current of the system is analyzed. To suppress the CM current resonance, an EMC quantitative design method is presented. Two Y capacitors are connected in parallel at the connection between the fast-charging cable and the bus capacitor to suppress the resonance. The design guideline of the Y capacitors is presented, and the influence on the leakage current of the system is considered. Conduction and radiation EMI tests are carried out for the all-in-one motor drive module to validate the effectiveness of the EMC quantitative design method.

The primary significance of this article lies in providing a solution to the complex problem of modeling and suppression of CM current resonance in an all-in-one motor drive module. Meantime, the proposed modeling method can be applied to solving the EMI noise coupling problems between all-in-one motor drive module and on-board electronic devices.

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